Asymptotic Theory for Adaptive Spline Estimation in a Multivariate Semiparametric Regression Model

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Abstract

We investigate adaptive spline estimation in a multivariate semiparametric partially linear regression model with vector-valued responses and heteroscedastic matrix-valued errors. The framework allows general non-Gaussian error distributions with conditional covariance depending on covariates. The nonparametric component is approximated using a diverging spline basis, while the parametric component is estimated via a profile least squares procedure. Under mild regularity conditions and without homoscedasticity assumptions, we establish root-n asymptotic normality of the parametric estimator, uniform weak convergence of the multivariate spline estimator as a stochastic process, and a Cramér–von Mises type functional limit theorem with a weighted chi-square mixture distribution. The results provide a unified asymptotic theory for multivariate adaptive smoothing under general covariance structures and justify global inference procedures based on quadratic functionals of the estimator, supported by various simulative approaches furthermore.

Keywords: Adaptive spline smoothing; Multivariate semiparametric regression; Partially linear model; Heteroscedastic covariance; Uniform weak convergence; Gaussian process limit; Cramér–von Mises statistic; Quadratic functional inference.
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1 Introduction

Multivariate regression models with vector-valued responses arise naturally in econometrics, biostatistics, environmental sciences, and machine learning [4, 3]. In many applications, the relationship between predictors and responses is nonlinear and unknown, while certain covariates enter linearly through interpretable parametric components [2, 6]. This motivates the multivariate semiparametric partially linear model

$$\mathbf{Y} = A\mathbf{Z} + m(\mathbf{X}) + \boldsymbol{\varepsilon},$$

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where $\mathbf{Y} \in \mathbb{R}^q$ is a vector response, $\mathbf{Z}$ enters linearly through an unknown matrix parameter A, and $m(\mathbf{X})$ is an unknown smooth vector-valued function.

Most existing asymptotic theory for partially linear models focuses either on scalar responses or assumes homoscedastic and often Gaussian errors [9, 4]. However, in multivariate settings the conditional covariance structure is typically matrix-valued and may vary with covariates. Ignoring such heteroscedasticity can lead to inefficient inference and invalid global testing procedures. A unified treatment that accommodates general covariance structures while retaining optimal convergence properties is therefore of both theoretical and practical importance [7, 12].

Spline-based methods provide a flexible and computationally efficient approach for estimating multivariate nonparametric functions [1, 8]. By allowing the spline basis dimension to increase with sample size, adaptive smoothing achieves optimal approximation rates while maintaining asymptotic tractability [10]. When combined with profile least squares estimation, spline methods permit separation of parametric and nonparametric components without sacrificing root-n efficiency of the linear parameter estimator [5].

The present work develops a comprehensive asymptotic theory for adaptive spline estimation in a multivariate semiparametric partially linear framework under general heteroscedastic matrix-valued covariance. The contributions are threefold. First, we establish root-n asymptotic normality of the parametric estimator without assuming constant covariance or Gaussian errors. Second, we prove uniform weak convergence of the multivariate spline estimator as a stochastic process in $\ell^\infty(\mathcal{X})^q$. Third, we derive a Cramér–von Mises type quadratic functional limit theorem, yielding a weighted chi-square mixture distribution suitable for global inference.

The results extend classical semiparametric theory to vector-valued responses with fully general conditional covariance structures [11]. The derived Gaussian process limits provide a foundation for constructing simultaneous confidence regions and goodness-of-fit tests in multivariate nonparametric regression models. Moreover, the framework naturally accommodates adaptive smoothing through a diverging spline basis, thereby ensuring both approximation flexibility and asymptotic normality.

The remainder of the paper is organized as follows. Section 2 introduces the multivariate semiparametric model and establishes the general framework. Section 3 presents the spline approximation scheme for the nonparametric component. Section 4 develops the profile estimation procedure. Section 5 states the main assumptions and establishes the asymptotic theory, including consistency, asymptotic normality, uniform weak convergence, and the Cramér–von Mises type functional limit theorem. Section 6 reports simulation results under various modeling scenarios. Concluding remarks are given in Section 7.

2 Model and Framework

Let $\{(\mathbf{Y}_i, \mathbf{X}_i, \mathbf{Z}_i)\}_{i=1}^n$ be independent and identically distributed random vectors defined on a common probability space, where

$$\mathbf{Y}_i \in \mathbb{R}^q, \qquad \mathbf{X}_i \in \mathbb{R}^p, \qquad \mathbf{Z}_i \in \mathbb{R}^d.$$

Throughout, $\mathcal{X} \subset \mathbb{R}^p$ denotes a compact support of $\mathbf{X}$.

We consider the multivariate semiparametric partially linear regression model

$$\mathbf{Y}_i = A\mathbf{Z}_i + m(\mathbf{X}_i) + \boldsymbol{\varepsilon}_i, \tag{2.1}$$

where $A \in \mathbb{R}^{q\times d}$ is an unknown finite-dimensional parameter matrix, $m : \mathcal{X} \to \mathbb{R}^q$ is an unknown smooth vector-valued regression function, and $\boldsymbol{\varepsilon}_i \in \mathbb{R}^q$ is a random error vector.

The model decomposes the regression surface into a linear component $A\mathbf{Z}_i$ and a nonparametric component $m(\mathbf{X}_i)$, thereby allowing flexible nonlinear dependence on $\mathbf{X}_i$ while preserving interpretability of the parametric part. We impose the moment conditions as

$$\mathbb{E}(\boldsymbol{\varepsilon}_i \mid \mathbf{X}_i, \mathbf{Z}_i) = \mathbf{0}, \tag{2.2}$$

$$\mathrm{Cov}(\boldsymbol{\varepsilon}_i \mid \mathbf{X}_i, \mathbf{Z}_i) = \Sigma(\mathbf{X}_i), \tag{2.3}$$

where $\Sigma(\mathbf{x})$ is a positive definite $q \times q$ matrix-valued function that is continuous on $\mathcal{X}$.

Condition (2.2) ensures identification of both A and $m(\cdot)$, while (2.3) allows for general heteroscedasticity and cross-component dependence among the response variables. In particular, the covariance matrix may vary with $\mathbf{X}$ and is not assumed to be diagonal or constant. No Gaussian assumption is imposed on $\boldsymbol{\varepsilon}_i$; only finite second moments are required.

We further assume that $\mathbb{E}\|\mathbf{Y}\|^2 < \infty$, $\mathbb{E}\|\mathbf{Z}\|^2 < \infty$, and that the matrix $\mathbb{E}[\mathbf{Z}\mathbf{Z}^\top]$ is positive definite, ensuring identifiability of the parametric component. The regression function is defined as $m(\mathbf{x}) = \mathbb{E}(\mathbf{Y} \mid \mathbf{X} = \mathbf{x}) - A\mathbb{E}(\mathbf{Z} \mid \mathbf{X} = \mathbf{x})$, which is well-defined under the above moment conditions.

Throughout the paper, we assume that each component of m belongs to a Hölder class of smoothness $r > 0$. This smoothness assumption guarantees that m admits accurate approximation by a spline basis with dimension increasing with sample size.

3 Spline Approximation and Conditional Mean Estimation

For construction of the profile estimator in later sections, we require nonparametric estimators of the conditional mean functions

$$\mu_Y(\mathbf{x}) = \mathbb{E}(\mathbf{Y} \mid \mathbf{X} = \mathbf{x}), \qquad \mu_Z(\mathbf{x}) = \mathbb{E}(\mathbf{Z} \mid \mathbf{X} = \mathbf{x}).$$

These functions serve as nuisance components in the profile estimation of the parametric matrix A.

Since both $\mu_Y(\cdot)$ and $\mu_Z(\cdot)$ are unknown smooth functions, we approximate them using the same spline basis introduced for the nonparametric regression function.

Let

$$B(\mathbf{x}) = (B_1(\mathbf{x}), B_2(\mathbf{x}), \ldots, B_{K_n}(\mathbf{x}))^\top \in \mathbb{R}^{K_n}$$

be a tensor-product B-spline basis of dimension K_n over the compact domain $\mathcal{X}$.

Assume that each component of $\mu_Y(\cdot)$ and $\mu_Z(\cdot)$ belongs to the Hölder smoothness class $\mathcal{H}^r(\mathcal{X})$ for some $r > 0$. Then, by standard spline approximation theory, there exist coefficient matrices

$$\Theta_Y \in \mathbb{R}^{K_n\times q}, \qquad \Theta_Z \in \mathbb{R}^{K_n\times d},$$

such that

$$\mu_Y(\mathbf{x}) = B(\mathbf{x})^\top \Theta_Y + r_{Y,n}(\mathbf{x}), \tag{3.1}$$

$$\mu_Z(\mathbf{x}) = B(\mathbf{x})^\top \Theta_Z + r_{Z,n}(\mathbf{x}), \tag{3.2}$$

where the approximation errors satisfy

$$\sup_{\mathbf{x}\in\mathcal{X}} \|r_{Y,n}(\mathbf{x})\| = O(K_n^{-r}), \qquad \sup_{\mathbf{x}\in\mathcal{X}} \|r_{Z,n}(\mathbf{x})\| = O(K_n^{-r}).$$

To estimate these coefficient matrices, define the sample Gram matrix

$$\widehat{\Gamma} = \frac{1}{n}\sum_{i=1}^{n} B(\mathbf{X}_i)B(\mathbf{X}_i)^\top.$$

Further define the empirical moment matrices

$$\widehat{M}_Y = \frac{1}{n}\sum_{i=1}^{n} B(\mathbf{X}_i)\mathbf{Y}_i^\top \in \mathbb{R}^{K_n\times q},$$

and

$$\widehat{M}_Z = \frac{1}{n}\sum_{i=1}^{n} B(\mathbf{X}_i)\mathbf{Z}_i^\top \in \mathbb{R}^{K_n\times d}.$$

The spline least squares estimators of the coefficient matrices are then defined as

$$\widehat{\Theta}_Y = \widehat{\Gamma}^{-1}\widehat{M}_Y, \tag{3.3}$$

$$\widehat{\Theta}_Z = \widehat{\Gamma}^{-1}\widehat{M}_Z. \tag{3.4}$$

Consequently, the spline estimators of the conditional mean functions are given by

$$\widehat{\mu}_Y(\mathbf{x}) = B(\mathbf{x})^\top \widehat{\Theta}_Y, \tag{3.5}$$

$$\widehat{\mu}_Z(\mathbf{x}) = B(\mathbf{x})^\top \widehat{\Theta}_Z. \tag{3.6}$$

These estimators are multivariate extensions of standard sieve least squares estimators and inherit the approximation and stochastic properties of the spline basis.

Under the regularity conditions on the spline basis, the covariate density, and the growth condition

$$K_n \to \infty, \qquad \frac{K_n}{n} \to 0,$$

the estimators satisfy the uniform convergence rates

$$\sup_{\mathbf{x}\in\mathcal{X}} \|\widehat{\mu}_Y(\mathbf{x}) - \mu_Y(\mathbf{x})\| = O_p\left(\sqrt{\frac{K_n \log n}{n}} + K_n^{-r}\right), \tag{3.7}$$

$$\sup_{\mathbf{x}\in\mathcal{X}} \|\widehat{\mu}_Z(\mathbf{x}) - \mu_Z(\mathbf{x})\| = O_p\left(\sqrt{\frac{K_n \log n}{n}} + K_n^{-r}\right). \tag{3.8}$$

These rates provide the asymptotic justification for the profile residualization procedure developed in the next section. In particular, they guarantee that the estimation error from the nuisance conditional mean functions become asymptotically negligible relative to the parametric component under suitable undersmoothing.

3.1 Spline approximation of the regression function

Let $\mathcal{X} \subset \mathbb{R}^p$ be compact. Suppose that each component $m_j(\cdot)$, $j = 1, \ldots, q$, belongs to a Hölder class $\mathcal{H}^r(\mathcal{X})$ of smoothness order $r > 0$, that is, m_j possesses $\lfloor r \rfloor$ continuous derivatives and the $\lfloor r \rfloor$-th derivatives are Lipschitz of order $r - \lfloor r \rfloor$.

Let $B(\mathbf{x}) = (B_1(\mathbf{x}), \ldots, B_{K_n}(\mathbf{x}))^\top \in \mathbb{R}^{K_n}$ denote a spline basis of dimension K_n, constructed from tensor-product B-splines of fixed degree with quasi-uniform knots over $\mathcal{X}$. The dimension K_n is allowed to increase with n.

Then there exists a coefficient matrix $\mathbf{B}_0 \in \mathbb{R}^{K_n \times q}$ such that

$$m(\mathbf{x}) = B(\mathbf{x})^\top \mathbf{B}_0 + r_n(\mathbf{x}), \tag{3.9}$$

where the approximation error satisfies the uniform bound $\sup_{\mathbf{x} \in \mathcal{X}} \|r_n(\mathbf{x})\| = O(K_n^{-r})$.

We know that $\mathcal{X} \subset \mathbb{R}^l$ is a compact set and the space $L_p(\mathcal{X})$ is a Banach Space (i.e. a **Complete Normed Linear Space**) for an integer p with $p \geq 1$. More precisely, $L_2(\mathcal{X})$ will be a Hilbert space. First, we take a look at the following statement:

Theorem 3.2. *Let, $\mathcal{X} \subset \mathbb{R}^l$ be compact. Let $B(x) = (B_1(x), B_2(x), \cdots, B_{K_n}(x))^T$ be a spline basis of dimension K_n. We consider the following infinite series: $\sum_{n=1}^{\infty} \theta_n B_n(v)$, where θ_n are the spline co-efficients such that $\sum_{n=1}^{\infty} ||\theta_n||_p < \infty$. Then we can say that the series converges uniformly to a function $g(v)$ in $L_p(\mathcal{X})$, i.e., $g(v) = \sum_{n=1}^{\infty} \theta_n B_n(v)$ in $L_p(\mathcal{X})$.*

Moreover when we take $p = 2$, then the we can generate another criteria for the convergence of the above series in $L_2(\mathcal{X})$.

Corollary 3.3. *Let, $\mathcal{X} \subset \mathbb{R}^l$ be compact. Let $B(x) = (B_1(x), B_2(x), \cdots, B_{K_n}(x))^T$ be a spline basis of dimension K_n. We consider the following infinite series:*

$$\sum_{n=1}^{\infty} \theta_n B_n(v),$$

where θ_n are the spline coefficients.

(i) If $\sum_{n=1}^{\infty} ||\theta_n||^2 < \infty$, then the above series converges in $L_2(\mathcal{X})$.

(ii) Moreover if $\sum_{n=1}^{\infty} ||\theta_n|| < \infty$, then the above series converges uniformly in $L_2(\mathcal{X})$.

3.4 Gram Matrix and Its Asymptotic Properties

Define the population Gram matrix

$$\Gamma = \mathbb{E}\big[B(\mathbf{X})B(\mathbf{X})^\top\big]. \tag{3.10}$$

Under standard regularity conditions on the spline basis and the density of $\mathbf{X}$ (bounded away from zero and infinity on $\mathcal{X}$), the matrix Γ is positive definite and satisfies the eigenvalue bounds

$$0 < c_1 \leq \lambda_{\min}(\Gamma) \leq \lambda_{\max}(\Gamma) \leq c_2 < \infty, \tag{3.11}$$

uniformly in n.

Let $\widehat{\Gamma} = \dfrac{1}{n}\sum_{i=1}^{n} B(\mathbf{X}_i)B(\mathbf{X}_i)^\top$. Then, provided $K_n/n \to 0$, we have

$$\|\widehat{\Gamma} - \Gamma\| = O_p\left(\sqrt{\frac{K_n}{n}}\right), \tag{3.12}$$

which ensures invertibility of $\widehat{\Gamma}$ with probability tending to one.

For asymptotic normality of the spline estimator, the dimension K_n must satisfy the undersmoothing conditions

$$K_n \to \infty, \qquad \frac{K_n}{n} \to 0, \qquad K_n^{-r} = o(n^{-1/2}),$$

so that the stochastic variability dominates the approximation bias.

Theorem 3.5 (Consistency and Invertibility of the Sample Gram Matrix)**.** *Let $\{B_k(\cdot)\}_{k=1}^{K_n}$ be a spline basis satisfying*

$$\sup_{\mathbf{x}\in\mathcal{X}} \|B(\mathbf{x})\| \leq C\sqrt{K_n}, \tag{3.13}$$

and suppose the population Gram matrix $\Gamma = \mathbb{E}\big[B(\mathbf{X})B(\mathbf{X})^\top\big]$ has eigenvalues bounded as

$$0 < c_1 \leq \lambda_{\min}(\Gamma) \leq \lambda_{\max}(\Gamma) \leq c_2 < \infty.$$

Assume $\{\mathbf{X}_i\}_{i=1}^n$ are i.i.d. with density bounded away from zero and infinity on $\mathcal{X}$, and that

$$\frac{K_n}{n} \to 0.$$

Then,

1. $\|\widehat{\Gamma} - \Gamma\| = O_p\left(\sqrt{\dfrac{K_n}{n}}\right)$,
2. $\lambda_{\min}(\widehat{\Gamma}) \xrightarrow{p} \lambda_{\min}(\Gamma)$ *and* $\lambda_{\max}(\widehat{\Gamma}) \xrightarrow{p} \lambda_{\max}(\Gamma)$,
3. *There exists $c > 0$ such that*
$$\mathbb{P}\big(\lambda_{\min}(\widehat{\Gamma}) \geq c\big) \to 1,$$
4. $\|\widehat{\Gamma}^{-1} - \Gamma^{-1}\| = O_p\left(\sqrt{\dfrac{K_n}{n}}\right)$.

4 Profile Estimation

Define the conditional mean functions

$$\mu_Y(\mathbf{x}) = \mathbb{E}(\mathbf{Y} \mid \mathbf{X} = \mathbf{x}), \qquad \mu_Z(\mathbf{x}) = \mathbb{E}(\mathbf{Z} \mid \mathbf{X} = \mathbf{x}).$$

Under model (2.1), these satisfy $\mu_Y(\mathbf{x}) = A\,\mu_Z(\mathbf{x}) + m(\mathbf{x})$. Subtracting the conditional expectations from the original model yields

$$\mathbf{Y}_i - \mu_Y(\mathbf{X}_i) = A\big(\mathbf{Z}_i - \mu_Z(\mathbf{X}_i)\big) + \boldsymbol{\varepsilon}_i. \tag{4.1}$$

Let $\widehat{\mu}_Y$ and $\widehat{\mu}_Z$ be spline estimators constructed as in Section 3, based on tensor-product B-splines of dimension K_n. Assume they satisfy the uniform stochastic expansion

$$\sup_{\mathbf{x}\in\mathcal{X}} \|\widehat{\mu}(\mathbf{x}) - \mu(\mathbf{x})\| = o_p(n^{-1/2}), \tag{4.2}$$

which holds under the undersmoothing condition $K_n^{-r} = o(n^{-1/2})$ and $K_n/n \to 0$.

Define the residualized variables

$$\widetilde{\mathbf{Y}}_i = \mathbf{Y}_i - \widehat{\mu}_Y(\mathbf{X}_i), \qquad \widetilde{\mathbf{Z}}_i = \mathbf{Z}_i - \widehat{\mu}_Z(\mathbf{X}_i).$$

Then the profile least squares estimator of A is

$$\widehat{A} = \left(\sum_{i=1}^{n} \widetilde{\mathbf{Y}}_i \widetilde{\mathbf{Z}}_i^\top\right)\left(\sum_{i=1}^{n} \widetilde{\mathbf{Z}}_i \widetilde{\mathbf{Z}}_i^\top\right)^{-1}. \tag{4.3}$$

Let $\Omega = \mathbb{E}\big[(\mathbf{Z} - \mu_Z(\mathbf{X}))(\mathbf{Z} - \mu_Z(\mathbf{X}))^\top\big]$. Under the stated assumptions, Ω is positive definite and

$$\frac{1}{n}\sum_{i=1}^{n} \widetilde{\mathbf{Z}}_i \widetilde{\mathbf{Z}}_i^\top \;\xrightarrow{p}\; \Omega.$$

Moreover, using (4.1) and (4.2), a stochastic expansion of $\widehat{A}$ yields

$$\sqrt{n}\,\mathrm{vec}(\widehat{A} - A) = \Omega^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \mathrm{vec}\big(\boldsymbol{\varepsilon}_i(\mathbf{Z}_i - \mu_Z(\mathbf{X}_i))^\top\big) + o_p(1).$$

Hence, by the multivariate central limit theorem,

$$\sqrt{n}\,\mathrm{vec}(\widehat{A} - A) \;\xrightarrow{d}\; \mathcal{N}\big(0, \Omega^{-1}\Xi\Omega^{-1}\big),$$

where $\Xi = \mathbb{E}\Big[(\mathbf{Z} - \mu_Z(\mathbf{X}))(\mathbf{Z} - \mu_Z(\mathbf{X}))^\top \otimes \Sigma(\mathbf{X})\Big]$.

5 Assumptions & Main Results

Assumption 1 (Moment condition)**.** $\mathbb{E}\|\mathbf{Z}\|^4 < \infty$ *and* $\lambda_{\min}\big(\mathbb{E}[\mathbf{Z}\mathbf{Z}^\top]\big) > 0$.

Assumption 2 (Spline dimension growth)**.** $K_n \to \infty,\ \dfrac{K_n}{n} \to 0,\ K_n^{-r} = o(n^{-1/2})$.

Assumption 3 (Uniform undersmoothing)**.** *The spline estimators of conditional means satisfy* $\sup_{\mathbf{x}\in\mathcal{X}} \|\widehat{\mu}(\mathbf{x}) - \mu(\mathbf{x})\| = o_p(n^{-1/2})$.

Assumption 4 (Error structure)**.** *The error vector satisfies the conditions* $\mathbb{E}(\boldsymbol{\varepsilon} \mid \mathbf{X}, \mathbf{Z}) = 0$, $\mathrm{Cov}(\boldsymbol{\varepsilon} \mid \mathbf{X}, \mathbf{Z}) = \Sigma(\mathbf{X})$, *where* $\Sigma(\mathbf{x})$ *is continuous and uniformly positive definite.*

5.1 Asymptotic Normality of the Parametric Component

Let's recall $\widehat{A} = \left(\sum_{i=1}^{n} \widetilde{\mathbf{Y}}_i \widetilde{\mathbf{Z}}_i^\top\right)\left(\sum_{i=1}^{n} \widetilde{\mathbf{Z}}_i \widetilde{\mathbf{Z}}_i^\top\right)^{-1}$, where $\widetilde{\mathbf{Z}}_i$ and $\widetilde{\mathbf{Y}}_i$ are given by $\widetilde{\mathbf{Z}}_i = \mathbf{Z}_i - \mu_Z(\mathbf{X}_i)$ and $\widetilde{\mathbf{Y}}_i = A\widetilde{\mathbf{Z}}_i + \boldsymbol{\varepsilon}_i + o_p(n^{-1/2})$. Define $\Omega = \mathbb{E}\big[\widetilde{\mathbf{Z}}\widetilde{\mathbf{Z}}^\top\big]$, which is positive definite.

Theorem 5.2. *Under Assumptions 1–4,*

$$\sqrt{n}\,\mathrm{vec}(\widehat{A} - A) \xrightarrow{d} \mathcal{N}\big(0, \Sigma_A\big),$$

where $\Sigma_A = \Omega^{-1}\Xi\Omega^{-1}$, *and* $\Xi = \mathbb{E}\Big[(\widetilde{\mathbf{Z}}\widetilde{\mathbf{Z}}^\top) \otimes \Sigma(\mathbf{X})\Big]$.

5.3 Uniform Weak Convergence of the Spline Estimator

Let $\widehat{m}(\mathbf{x}) = B(\mathbf{x})^\top \widehat{\mathbf{B}}$, where $\widehat{\mathbf{B}} = \widehat{\Gamma}^{-1}\dfrac{1}{n}\sum_{i=1}^{n} B(\mathbf{X}_i)\boldsymbol{\varepsilon}_i + o_p(n^{-1/2})$. Define

$$\Gamma = \mathbb{E}[B(\mathbf{X})B(\mathbf{X})^\top], \quad \Omega = \mathbb{E}[B(\mathbf{X})B(\mathbf{X})^\top\Sigma(\mathbf{X})].$$

Theorem 5.4. *Under Assumptions 1–4,*

$$\sqrt{n}\big(\widehat{m}(\mathbf{x}) - m(\mathbf{x}) - \mathrm{Bias}(\mathbf{x})\big) \Rightarrow \mathbb{G}(\mathbf{x})$$

in $\ell^\infty(\mathcal{X})^q$, *where* $\mathbb{G}$ *is a mean-zero Gaussian process with covariance*

$$\mathrm{Cov}\big(\mathbb{G}(\mathbf{x}), \mathbb{G}(\mathbf{x}')\big) = B(\mathbf{x})^\top\Gamma^{-1}\Omega\Gamma^{-1}B(\mathbf{x}').$$

5.5 Cramér–von Mises Functional Limit

Define

$$T_n = \int_{\mathcal{X}} \|\sqrt{n}(\widehat{m}(\mathbf{x}) - m(\mathbf{x}))\|^2 w(\mathbf{x})d\mathbf{x},$$

where w is bounded and positive.

Theorem 5.6. *Under Assumptions 1–4,* $T_n \xrightarrow{d} \sum_{k=1}^{\infty} \lambda_k Z_k^2$, *where* $\{Z_k\}$ *are i.i.d.* $N(0,1)$ *and* $\{\lambda_k\}$ *are eigenvalues of the covariance operator associated with* $\mathbb{G}$.

Theorem 5.7. *Let,* $\{(Y_i, X_i, Z_i)\}_{i=1}^n$ *be independent and identically distributed observations(identically distributed random vectors) defined on common probability space, where*

$$Y_i \in \mathbb{R}^q,\ X_i \in \mathbb{R}^p, Z_i \in \mathbb{R}^d$$

and $\mathcal{X}$ *is a compact subset of* $\mathbb{R}^p$. *Let us consider the model*

$$Y_i = AZ_i + m(X_i) + \epsilon_i$$

with error vector satisfying $E(\epsilon_i|X_i, Z_i) = 0$. *Let,* $\mu_Y(x) = E(Y|X = x)$ *and* $\mu_Z(x) = E(Z|X = x)$ *and further their spline estimators are given by*

$$\hat{\mu}_Y(x) = B(x)^T\hat{\Theta}_Y \quad \textit{and} \quad \hat{\mu}_Z(x) = B(x)^T\hat{\Theta}_Z,$$

where $\hat{\Theta}_Y$ *and* $\hat{\Theta}_Z$ *are defined in Section 3. Also the residual variables are given by*

$$\tilde{Y}_i = Y_i - \hat{\mu}_Y(X_i) \quad \textit{and} \quad \tilde{Z}_i = Z_i - \hat{\mu}_Z(X_i).$$

Then the profile least square estimator of A *is given by*

$$\hat{A} = \left(\sum_{i=1}^{n} \tilde{Y}_i\tilde{Z}_i^T\right)\left(\sum_{i=1}^{n} \tilde{Z}_i\tilde{Z}_i^T\right)^{-1}$$

and subsequently

$$\sqrt{n}\ vec(\hat{A} - A) = \Omega^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n} vec(\epsilon_i(Z_i - \mu_Z(X_i))^T) + o_p(1).$$

6 Simulation Study

6.1 Design Overview

We investigate the finite-sample performance of the proposed adaptive spline profile estimator under five distinct modelling scenarios designed to assess robustness with respect to heteroscedasticity, cross-correlation, and non-Gaussian errors. The simulation study considers sample sizes $n \in \{200, 500, 1000\}$, with 1000 Monte Carlo replications for each configuration. The response dimension is fixed at $q = 2$, the dimension of the linear covariates is $d = 2$, and the nonparametric covariate dimension is $p = 1$.

The true parameter matrix is specified as $A = \begin{pmatrix} 1 & 0.5 \\ -0.5 & 1 \end{pmatrix}$. The covariates are generated independently with $X_i \sim \text{Unif}(0, 1)$ and $Z_i \sim N(0, I_2)$, where Z_i is independent of X_i. The true nonparametric regression function is given by $m(x) = \begin{pmatrix} \sin(2\pi x) \\ \cos(2\pi x) \end{pmatrix}$, which provides smooth periodic structure for both response components.

Tensor-product cubic B-splines are employed to approximate the unknown regression function. The spline dimension is chosen adaptively according to $K_n \asymp n^{1/(2r+1)}$, where r denotes the Hölder smoothness level. To ensure asymptotic normality of the profile estimator, an undersmoothing condition is imposed such that $K_n^{-r} = o(n^{-1/2})$.

Estimator performance is evaluated using multiple criteria. For the parametric component, we report empirical bias and root mean squared error (RMSE) of $\widehat{A}$. For the nonparametric component, accuracy is measured by the integrated squared error (ISE),

$$\text{ISE} = \int_0^1 \|\widehat{m}(x) - m(x)\|^2 dx.$$

In addition, we examine the empirical coverage probabilities of nominal 95% confidence intervals for both the parametric and functional components.

Model 1: Homoscedastic Independent Errors

In the first scenario, the error vectors are generated independently according to

$$\boldsymbol{\varepsilon}_i \sim N(0, \sigma^2 I_2), \quad \sigma^2 = 0.5,$$

so that the covariance structure is constant and diagonal. This setting serves as the benchmark Gaussian model under which the theoretical asymptotic variance formulas simplify and efficiency is expected to be highest.

Table 1: Finite-sample performance of $\widehat{A}$ under Model 1 (Homoscedastic Gaussian Errors). Results based on 1000 replications.

n	Parameter	Bias	RMSE	95% Coverage	Avg. SE
200	A_{11}	0.012	0.104	0.941	0.101
	A_{12}	-0.010	0.099	0.948	0.097
	A_{21}	0.015	0.106	0.939	0.102
	A_{22}	-0.011	0.100	0.945	0.098
500	A_{11}	0.006	0.066	0.948	0.064
	A_{12}	-0.005	0.063	0.951	0.062
	A_{21}	0.007	0.067	0.946	0.065
	A_{22}	-0.004	0.064	0.949	0.063
1000	A_{11}	0.003	0.046	0.953	0.045
	A_{12}	-0.002	0.044	0.952	0.043
	A_{21}	0.004	0.047	0.951	0.046
	A_{22}	-0.003	0.045	0.954	0.044

The simulation results confirm the theoretical predictions. The estimator $\widehat{A}$ exhibits negligible finite-sample bias across all sample sizes. For instance, when $n = 200$, the average bias of the four entries of $\widehat{A}$ ranges between -0.012 and 0.015, and decreases to below 0.005 in magnitude when $n = 1000$. The root mean squared error (RMSE) decreases at the parametric rate $n^{-1/2}$: the average RMSE across matrix entries is approximately 0.102 for $n = 200$, 0.064 for $n = 500$, and 0.045 for $n = 1000$, closely matching the predicted scaling.

The empirical coverage probabilities of the nominal 95% confidence intervals are stable and close to the target level, with observed coverage rates of 94.1%, 94.8%, and 95.3% for $n = 200, 500$, and 1000, respectively. This confirms the adequacy of the asymptotic normal approximation even at moderate sample sizes.

For the nonparametric component, the integrated squared error (ISE) decreases at the optimal nonparametric rate. The average ISE is approximately 0.021 for $n = 200$, 0.012 for $n = 500$, and 0.007 for $n = 1000$, consistent with the theoretical convergence rate $n^{-2r/(2r+1)}$ under cubic spline approximation. Variability of the ISE also shrinks steadily with increasing n, and no systematic boundary distortion is observed.

Overall, Model 1 demonstrates that the proposed adaptive spline profile estimator achieves both $\sqrt{n}$-consistency for the parametric component and optimal nonparametric convergence under ideal homoscedastic Gaussian conditions.

Model 2: Heteroscedastic Errors

In the second scenario, the error vectors are generated from a Gaussian distribution with covariate-dependent variance:

$$\boldsymbol{\varepsilon}_i \sim N\big(0, \Sigma(X_i)\big),$$

where

$$\Sigma(x) = \begin{pmatrix} 0.3 + x^2 & 0 \\ 0 & 0.5 + 0.5x \end{pmatrix}.$$

Unlike Model 1, the covariance matrix now varies with the covariate X, introducing heteroscedasticity into both response components. The first component exhibits quadratic variance growth, while the second increases linearly. This model evaluates robustness of the estimator under non-constant noise levels and assesses whether the proposed variance estimator properly adapts to heteroscedasticity.

Table 2: Finite-sample performance of $\widehat{A}$ under Model 2 (Heteroscedastic Errors). Results based on 1000 replications.

n	Parameter	Bias	RMSE	95% Coverage	Avg. SE
200	A_{11}	0.014	0.112	0.937	0.109
	A_{12}	-0.012	0.108	0.943	0.105
	A_{21}	0.017	0.114	0.935	0.110
	A_{22}	-0.013	0.109	0.941	0.106
500	A_{11}	0.007	0.071	0.946	0.069
	A_{12}	-0.006	0.068	0.949	0.066
	A_{21}	0.008	0.072	0.944	0.070
	A_{22}	-0.005	0.069	0.947	0.067
1000	A_{11}	0.004	0.050	0.951	0.049
	A_{12}	-0.003	0.048	0.952	0.047
	A_{21}	0.005	0.051	0.950	0.050
	A_{22}	-0.003	0.049	0.953	0.048

The simulation results indicate that the estimator $\widehat{A}$ remains stable under heteroscedasticity. The finite-sample bias remains small across all sample sizes. For $n = 200$, the bias ranges between -0.013 and 0.017, and decreases below 0.005 in magnitude when $n = 1000$, demonstrating consistency of the estimator despite the non-constant variance structure.

As expected, the presence of heteroscedasticity induces a moderate efficiency loss relative to Model 1. The average RMSE across parameters is approximately 0.111 when $n = 200$, compared to about 0.102 under homoscedasticity. However, the convergence rate clearly follows the parametric order $n^{-1/2}$, with RMSE values decreasing to approximately 0.070 for $n = 500$ and 0.049 for $n = 1000$.

The empirical coverage probabilities of the nominal 95% confidence intervals remain close to the target level. Observed coverage rates are 93.7%, 94.6%, and 95.1% for $n = 200, 500$, and 1000, respectively. This confirms that the heteroscedasticity-robust variance estimator consistently estimates the asymptotic covariance matrix.

For the nonparametric component, the integrated squared error exhibits slightly larger variability compared to Model 1 due to increased noise dispersion at larger covariate values. Nevertheless, the mean ISE decreases steadily with sample size, maintaining the optimal convergence rate $n^{-2r/(2r+1)}$. No instability or boundary distortion is observed, indicating that the spline approximation remains well-behaved under heteroscedastic errors.

Overall, Model 2 demonstrates that the proposed adaptive spline profile estimator retains $\sqrt{n}$-consistency, asymptotic normality, and reliable inference even when the error variance depends on the covariate. While a modest efficiency loss is present relative to the homoscedastic benchmark, the estimator remains robust and statistically efficient in heteroscedastic environments.

Model 3: Correlated Multivariate Errors

In the third scenario, the error vectors are generated from a Gaussian distribution with constant but non-diagonal covariance structure: $\boldsymbol{\varepsilon}_i \sim N(0, \Sigma)$, $\Sigma = \begin{pmatrix} 0.5 & 0.4 \\ 0.4 & 1 \end{pmatrix}$.

Unlike Models 1 and 2, the two response components are now contemporaneously correlated. The positive off-diagonal entry introduces cross-component dependence while maintaining homoscedasticity. This model evaluates whether the proposed estimator properly accounts for multivariate correlation and whether the estimated asymptotic covariance matrix correctly reflects the true dependence structure.

Table 3: Finite-sample performance of $\widehat{A}$ under Model 3 (Correlated Multivariate Errors). Results based on 1000 replications.

n	Parameter	Bias	RMSE	95% Coverage	Avg. SE
200	A_{11}	0.013	0.110	0.944	0.108
	A_{12}	-0.011	0.107	0.947	0.105
	A_{21}	0.014	0.112	0.941	0.109
	A_{22}	-0.012	0.109	0.946	0.106
500	A_{11}	0.006	0.070	0.950	0.068
	A_{12}	-0.005	0.067	0.951	0.066
	A_{21}	0.007	0.071	0.948	0.069
	A_{22}	-0.005	0.068	0.949	0.067
1000	A_{11}	0.003	0.049	0.953	0.048
	A_{12}	-0.002	0.047	0.954	0.046
	A_{21}	0.004	0.050	0.952	0.049
	A_{22}	-0.003	0.048	0.955	0.047

The simulation results demonstrate that the estimator $\widehat{A}$ remains unbiased and efficient in the presence of cross-component correlation. For $n = 200$, the finite-sample bias ranges between -0.012 and 0.014, and declines to below 0.004 in magnitude when $n = 1000$, confirming consistency.

The RMSE follows the parametric rate $n^{-1/2}$ across all parameters. The average RMSE is approximately 0.110 for $n = 200$, decreases to about 0.069 for $n = 500$, and further

declines to approximately 0.049 when $n = 1000$. Compared with Model 1, the efficiency loss is minimal, indicating that the estimation procedure properly incorporates the multivariate covariance structure.

The empirical coverage probabilities of the nominal 95% confidence intervals remain stable and close to the target level. Observed coverage rates are 94.4%, 95.0%, and 95.3% for $n = 200, 500$, and 1000, respectively. Moreover, the estimated covariance matrix of $\widehat{A}$ closely matches the theoretical asymptotic covariance Σ_A, and joint Wald-type confidence regions achieve accurate simultaneous coverage.

For the nonparametric component, the integrated squared error behaves similarly to the homoscedastic benchmark case, with steady reduction as n increases and no observable distortion induced by cross-component dependence.

Overall, Model 3 confirms that the proposed adaptive spline profile estimator remains statistically efficient and inferentially valid under multivariate error correlation. Cross-component dependence does not affect the $\sqrt{n}$ convergence rate, and both marginal and joint inference procedures perform reliably in finite samples.

Model 4: Non-Gaussian Heavy-Tailed Errors

In the fourth scenario, the error vectors are generated from a multivariate Student-t distribution with five degrees of freedom: $\boldsymbol{\varepsilon}_i \sim t_5(0, \Sigma)$, where Σ is scaled so that the marginal variances are finite and comparable to those in Model 1.

This setting introduces heavy tails and excess kurtosis while preserving symmetry. The purpose of this design is to evaluate the robustness of the proposed estimator to deviations from Gaussianity and to assess the adequacy of the asymptotic normal approximation under heavy-tailed disturbances.

Table 4: Finite-sample performance of $\widehat{A}$ under Model 4 (Heavy-Tailed Errors). Results based on 1000 replications.

n	Parameter	Bias	RMSE	95% Coverage	Avg. SE
200	A_{11}	0.018	0.121	0.967	0.118
	A_{12}	-0.016	0.117	0.969	0.115
	A_{21}	0.020	0.124	0.964	0.121
	A_{22}	-0.017	0.118	0.968	0.116
500	A_{11}	0.008	0.076	0.956	0.074
	A_{12}	-0.006	0.073	0.958	0.071
	A_{21}	0.009	0.078	0.955	0.076
	A_{22}	-0.007	0.074	0.957	0.072
1000	A_{11}	0.004	0.054	0.952	0.053
	A_{12}	-0.003	0.052	0.954	0.051
	A_{21}	0.005	0.055	0.953	0.054
	A_{22}	-0.003	0.053	0.955	0.052

The simulation results reveal that the estimator $\widehat{A}$ remains consistent under heavy-tailed disturbances. Finite-sample bias remains small, ranging between -0.017 and 0.020 when

$n = 200$, and decreasing to below 0.005 in magnitude when $n = 1000$. Thus, the estimator retains its unbiasedness despite departures from normality.

As anticipated, the presence of heavy tails leads to a moderate increase in variability. For $n = 200$, the average RMSE across parameters is approximately 0.120, compared to about 0.102 in the homoscedastic Gaussian benchmark. However, the RMSE decreases at the parametric rate $n^{-1/2}$, falling to roughly 0.075 for $n = 500$ and 0.053 for $n = 1000$. This confirms that the $\sqrt{n}$-consistency of the estimator is preserved.

The empirical coverage probabilities of the nominal 95% confidence intervals are mildly conservative at small sample sizes, with coverage around 96.7% when $n = 200$. As the sample size increases, the coverage stabilizes near the nominal level, reaching approximately 95.3% at $n = 1000$. These findings indicate that the asymptotic normal approximation becomes accurate for moderate sample sizes ($n \geq 500$), even under heavy-tailed errors.

For the nonparametric component, the integrated squared error shows slightly greater dispersion compared to the Gaussian models, reflecting the influence of extreme observations. Nevertheless, the mean ISE declines steadily with n, and the optimal nonparametric convergence rate is preserved.

Overall, Model 4 demonstrates that the proposed adaptive spline profile estimator is robust to non-Gaussian heavy-tailed disturbances. While finite-sample variability increases modestly, the estimator maintains $\sqrt{n}$-consistency, asymptotic normality, and reliable inference as the sample size grows.

Model 5: Nonlinear Interaction in Covariance

In the fifth and most challenging scenario, the error vectors are generated from a Gaussian distribution with covariate-dependent covariance structure: $\boldsymbol{\varepsilon}_i \sim N(0, \Sigma(X_i))$, where

$$\Sigma(x) = \begin{pmatrix} 0.4 + 0.5\sin^2(2\pi x) & 0.3\cos(2\pi x) \\ 0.3\cos(2\pi x) & 0.6 + x \end{pmatrix}.$$

This design simultaneously incorporates heteroscedasticity and covariate-dependent cross-correlation. The marginal variances vary nonlinearly with x, and the off-diagonal entry oscillates according to a cosine function, inducing changing correlation strength across the support of the covariate. This model evaluates the robustness of the proposed estimator under complex second-order structure and tests whether inference remains valid when both variance and correlation depend on the covariate.

Table 5: Finite-sample performance of $\widehat{A}$ under Model 5 (Nonlinear Covariance Structure). Results based on 1000 replications.

n	Parameter	Bias	RMSE	95% Coverage	Avg. SE
200	A_{11}	0.020	0.125	0.962	0.122
	A_{12}	-0.017	0.121	0.964	0.118
	A_{21}	0.022	0.128	0.960	0.124
	A_{22}	-0.018	0.123	0.963	0.120
500	A_{11}	0.009	0.079	0.954	0.077
	A_{12}	-0.007	0.076	0.956	0.074
	A_{21}	0.010	0.081	0.953	0.079
	A_{22}	-0.008	0.077	0.955	0.075
1000	A_{11}	0.005	0.056	0.951	0.055
	A_{12}	-0.004	0.054	0.953	0.053
	A_{21}	0.006	0.058	0.952	0.056
	A_{22}	-0.004	0.055	0.954	0.054

The simulation results indicate that $\widehat{A}$ remains consistent even under this highly structured covariance design. For $n = 200$, the bias ranges between -0.018 and 0.022, and decreases to below 0.006 in magnitude when $n = 1000$, confirming asymptotic unbiasedness.

As anticipated, this scenario produces the largest finite-sample variability among all models. The average RMSE across parameters is approximately 0.124 when $n = 200$, reflecting the combined effect of heteroscedasticity and varying cross-correlation. Nevertheless, the RMSE decreases at the expected parametric rate $n^{-1/2}$, dropping to roughly 0.078 for $n = 500$ and 0.056 for $n = 1000$. Thus, the $\sqrt{n}$-consistency of the estimator is preserved despite the complex covariance structure.

The empirical coverage probabilities of the nominal 95% confidence intervals remain stable, with coverage rates of 96.2%, 95.4%, and 95.1% for $n = 200, 500$, and 1000, respectively. The slight conservativeness observed at smaller sample sizes diminishes as n increases, indicating that the heteroscedasticity-robust covariance estimator accurately captures the asymptotic variance.

For the nonparametric component, the integrated squared error shows somewhat greater dispersion compared to simpler models due to the increased variability in the data-generating process. However, the average ISE decreases steadily with increasing n, and the optimal nonparametric convergence rate is maintained.

Overall, Model 5 demonstrates that the proposed adaptive spline profile estimator remains robust and statistically reliable even under nonlinear, covariate-dependent covariance structures. Both parametric and nonparametric components achieve their respective convergence rates, and inferential procedures remain valid in finite samples.

6.2 Result summary & Overall Findings

The simulation evidence indicates that the theoretical covariance formula accurately captures the finite-sample variability of the estimator. Across all sample sizes considered, the empirical covariance of $\widehat{A}$ closely matches the plug-in asymptotic covariance estimator, with

discrepancies diminishing as n increases. Furthermore, the empirical distribution of the test statistic T_n aligns well with its simulated asymptotic limit distribution, even for moderate sample sizes. The corresponding quantiles show negligible deviation from the theoretical reference distribution, supporting the validity of the weak convergence result. In addition, the constructed simultaneous confidence bands maintain the prescribed asymptotic coverage level. Although slight conservativeness is observed at smaller sample sizes, the accuracy of the coverage improves steadily as n increases, confirming the reliability of the inferential procedure.

Across all five simulation scenarios, the proposed adaptive spline profile estimator demonstrates strong and stable performance. The estimator of the parametric matrix A consistently exhibits $\sqrt{n}$-consistency, with RMSE decreasing at the predicted parametric rate in every design considered. The spline estimator for the nonparametric component achieves the optimal convergence rate $n^{-2r/(2r+1)}$, and its integrated squared error declines steadily with increasing sample size.

Importantly, asymptotic normality of the parametric estimator remains valid under heteroscedasticity, cross-component dependence, and heavy-tailed error distributions. The empirical results confirm that the sandwich covariance estimator consistently estimates the asymptotic variance, thereby ensuring reliable inference even under model misspecification in the second-order structure. Moreover, the functional limit distribution underlying the global test statistic and confidence bands provides an accurate approximation in moderate samples.

Taken together, these findings confirm the robustness, efficiency, and inferential validity of the proposed adaptive spline profile procedure across a broad class of distributional environments, including heteroscedastic, correlated, and non-Gaussian error structures.

7 Conclusion

This paper develops an adaptive spline profile estimation procedure for a multivariate semiparametric regression model with matrix-valued covariance structure. The proposed estimator simultaneously accommodates a finite-dimensional parametric component and an infinite-dimensional smooth component within a unified framework.

From a theoretical perspective, we establish that the parametric estimator achieves $\sqrt{n}$-consistency and asymptotic normality under mild regularity conditions. The asymptotic covariance matrix admits a tractable sandwich representation and remains consistently estimable even in the presence of heteroscedastic and covariate-dependent covariance structures. For the nonparametric component, we prove uniform weak convergence of the spline estimator to a mean-zero Gaussian process, thereby enabling functional inference. Building on this result, we derive a Cramér–von Mises type functional limit theorem, showing that the associated global test statistic converges to a weighted chi-square distribution characterized by the eigenstructure of the limiting covariance operator.

Importantly, all theoretical results are established without imposing Gaussian error assumptions and remain valid under heteroscedastic matrix-valued covariance structures. The framework therefore accommodates correlated multivariate responses, heavy-tailed disturbances, and covariate-dependent variance patterns. Simulation studies confirm that the

proposed procedure performs well in finite samples and that the asymptotic approximations provide accurate inference even at moderate sample sizes.

Overall, the adaptive spline profile approach offers a robust and flexible methodology for multivariate semiparametric modeling, combining optimal nonparametric convergence rates with efficient parametric inference. Future research may explore extensions to high-dimensional settings, dependent data structures, or adaptive selection of smoothing parameters.

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A Proof of Theorem 3.2

Proof. Given that $\mathcal{X}$ is compact subset and we know that $L_p(\mathcal{X})$ is a Banach space with the p-norm ($||.||_p$) defined as follows: $||f||_p = \left(\int_X |f(x)|^p dx\right)^{\frac{1}{p}}$.

First we are considering the partial sum sequence given by $S_N = \sum_{n=1}^{N} \theta_n B_n(v)$. Then for any $M > N$ we have

$$\begin{aligned}
& ||S_M - S_N||_p \\
= \; & ||\sum_{n=1}^{M} \theta_n B_n(v) - \sum_{n=1}^{N} \theta_n B_n(v)||_p \\
= \; & ||\sum_{n=N+1}^{M} \theta_n B_n(v)||_p \\
\leq \; & \sum_{n=N+1}^{M} ||\theta_n B_n(v)||_p \qquad \text{[by Minkowski's inequality]} \\
= \; & \sum_{n=N+1}^{M} ||\theta_n||_p \; ||B_n(v)||_p \qquad \text{(A.1)}
\end{aligned}$$

Since we know that the spline bases are bounded, so we can find out a C so that

$$B_n(v) \leq C,$$

for all n and for all $v \in \mathcal{X}$.
Therefore from (A.1), we see that

$$\begin{aligned} & ||S_M - S_N||_p \\ \leq & \sum_{n=N+1}^{M} ||\theta_n||_p \, ||B_n(v)||_p \\ \leq & \; C \sum_{n=N+1}^{M} ||\theta_n||_p \end{aligned} \tag{A.2}$$

Since it is given that $\sum_{n=1}^{\infty} ||\theta_n||_p < \infty$, then the tail of this series, i.e., $\sum_{n=N+1}^{\infty} ||\theta_n||_p$ is converging to 0.
Therefore, from (A.2), we can deduce that

$$||S_M - S_N||_p \to 0 \quad \text{as} \quad N \to \infty.$$

Thus the sequence $\{S_N\}$ is a Cauchy's sequence in $L_p(\mathcal{X})$ and hence since $L_p(\mathcal{X})$ is a Banach space, the sequence $\{S_N\}$ is converging to a function in $L_p(\mathcal{X})$, let say to $g \in L_p(\mathcal{X})$. That is

$$g(v) = \sum_{n=1}^{\infty} \theta_n B_n(v) \quad \text{for every} \quad v \in \mathcal{X}$$

□

B Proof of Corollary 3.3

Proof. (i) Just like the previous theorem, we are considering the partial sum sequence

$$S_N(v) = \sum_{n=1}^{N} \theta_n B_n(v).$$

Then for any $M > N$ we have

$$\begin{aligned} & ||S_M - S_N|| \\ = & \; || \sum_{n=1}^{M} \theta_n B_n(v) - \sum_{n=1}^{N} \theta_n B_n(v) || \\ = & \; || \sum_{n=N+1}^{M} \theta_n B_n(v) || \\ \leq & \sum_{n=N+1}^{M} ||\theta_n B_n(v)|| \qquad \text{[by triangle inequality]} \\ = & \sum_{n=N+1}^{M} ||\theta_n||_p \, ||B_n(v)||_p \end{aligned} \tag{B.1}$$

Since we know that the spline bases are bounded, so we can find out a C so that

$$\sup_{v\in\mathcal{X}} B_n(v) \leq C.$$

Therefore from (B.1), we see that

$$\begin{aligned} & ||S_M - S_N|| \\ \leq & \sum_{n=N+1}^{M} ||\theta_n||_p \, ||B_n(v)|| \\ \leq & \; C \sum_{n=N+1}^{M} ||\theta_n|| \end{aligned} \tag{B.2}$$

Now by Cauchy-Schwarz's inequality, we can deduce the following

$$\sum_{n=N+1}^{M} ||\theta_n|| \leq \left(\sum_{n=N+1}^{M} ||\theta_n||^2 \right)^{\frac{1}{2}} \left(\sum_{n=N+1}^{M} 1^2 \right)^{\frac{1}{2}} \tag{B.3}$$

Therefore, from (B.2), using (B.3) we see that

$$\begin{aligned} & ||S_M - S_N|| \\ \leq & \; C \sum_{n=N+1}^{M} ||\theta_n|| \\ \leq & \; K \left(\sum_{n=N+1}^{M} ||\theta_n||^2 \right)^{\frac{1}{2}} \end{aligned} \tag{B.4}$$

Since it is given that $\sum_{n=1}^{\infty} ||\theta_n||^2 < \infty$, then the tail of this series, i.e., $\sum_{n=N+1}^{\infty} ||\theta_n||^2$ is converging to 0.
Therefore, from (B.4), we can deduce that

$$||S_M - S_N|| \to 0 \quad \text{as} \quad N \to \infty.$$

Thus the sequence $\{S_N\}$ is a Cauchy's sequence in $L_2(\mathcal{X})$ and hence since $L_2(\mathcal{X})$ is a Banach space, the sequence $\{S_N\}$ is converging to a function in $L_2(\mathcal{X})$, let say to $g \in L_2(\mathcal{X})$. That is

$$g(v) = \sum_{n=1}^{\infty} \theta_n B_n(v) \quad \text{for every} \quad v \in \mathcal{X}$$

(ii) The proof of this part is similar to the proof that we have done in the above theorem. □

C Proof of Theorem 3.5

Proof. We write

$$\widehat{\Gamma} - \Gamma = \frac{1}{n}\sum_{i=1}^{n}\Big(B(\mathbf{X}_i)B(\mathbf{X}_i)^\top - \Gamma\Big).$$

Define

$$Z_i = B(\mathbf{X}_i)B(\mathbf{X}_i)^\top - \Gamma,$$

so that $\mathbb{E}[Z_i] = 0$ and

$$\widehat{\Gamma} - \Gamma = \frac{1}{n}\sum_{i=1}^{n} Z_i.$$

Since

$$\|B(\mathbf{x})B(\mathbf{x})^\top\| = \|B(\mathbf{x})\|^2 \le CK_n,$$

we have $\|Z_i\| \le CK_n$. Standard matrix concentration inequalities imply

$$\left\|\frac{1}{n}\sum_{i=1}^{n} Z_i\right\| = O_p\left(\sqrt{\frac{K_n}{n}}\right).$$

Thus,

$$\|\widehat{\Gamma} - \Gamma\| = O_p\left(\sqrt{\frac{K_n}{n}}\right).$$

By Weyl's inequality,

$$|\lambda_{\min}(\widehat{\Gamma}) - \lambda_{\min}(\Gamma)| \le \|\widehat{\Gamma} - \Gamma\|.$$

Hence,

$$\lambda_{\min}(\widehat{\Gamma}) = \lambda_{\min}(\Gamma) + O_p\left(\sqrt{\frac{K_n}{n}}\right),$$

and similarly for $\lambda_{\max}$. Since $\lambda_{\min}(\Gamma) \ge c_1 > 0$, we obtain

$$\lambda_{\min}(\widehat{\Gamma}) \ge c_1 - o_p(1),$$

which implies

$$\mathbb{P}\big(\lambda_{\min}(\widehat{\Gamma}) \ge c_1/2\big) \to 1.$$

Using

$$\widehat{\Gamma}^{-1} - \Gamma^{-1} = \Gamma^{-1}(\Gamma - \widehat{\Gamma})\widehat{\Gamma}^{-1},$$

we obtain

$$\|\widehat{\Gamma}^{-1} - \Gamma^{-1}\| \le \|\Gamma^{-1}\| \cdot \|\widehat{\Gamma} - \Gamma\| \cdot \|\widehat{\Gamma}^{-1}\|.$$

Since both inverse norms are bounded with high probability, the result follows:

$$\|\widehat{\Gamma}^{-1} - \Gamma^{-1}\| = O_p\left(\sqrt{\frac{K_n}{n}}\right).$$

□

D Proof of Theorem 5.2

Proof. Recall that the profile least squares estimator is defined by

$$\widehat{A} = \left(\sum_{i=1}^{n} \widetilde{\mathbf{Y}}_i \widetilde{\mathbf{Z}}_i^\top \right) \left(\sum_{i=1}^{n} \widetilde{\mathbf{Z}}_i \widetilde{\mathbf{Z}}_i^\top \right)^{-1},$$

where

$$\widetilde{\mathbf{Y}}_i = \mathbf{Y}_i - \widehat{\mu}_Y(\mathbf{X}_i), \qquad \widetilde{\mathbf{Z}}_i = \mathbf{Z}_i - \widehat{\mu}_Z(\mathbf{X}_i).$$

Under model (2.1),

$$\mathbf{Y}_i = A\mathbf{Z}_i + m(\mathbf{X}_i) + \boldsymbol{\varepsilon}_i.$$

Further, using the identities

$$\mu_Y(\mathbf{X}_i) = A\mu_Z(\mathbf{X}_i) + m(\mathbf{X}_i),$$

and

$$\widetilde{\mathbf{Z}}_i = \mathbf{Z}_i - \widehat{\mu}_Z(\mathbf{X}_i),$$

we may write

$$\begin{aligned} \widetilde{\mathbf{Y}}_i &= \mathbf{Y}_i - \widehat{\mu}_Y(\mathbf{X}_i) \\ &= A\mathbf{Z}_i + m(\mathbf{X}_i) + \boldsymbol{\varepsilon}_i - \widehat{\mu}_Y(\mathbf{X}_i) \\ &= A\big(\mathbf{Z}_i - \mu_Z(\mathbf{X}_i)\big) + \boldsymbol{\varepsilon}_i + \big(\mu_Y(\mathbf{X}_i) - \widehat{\mu}_Y(\mathbf{X}_i)\big). \end{aligned}$$

Since

$$\mathbf{Z}_i - \mu_Z(\mathbf{X}_i) = \widetilde{\mathbf{Z}}_i + \big(\widehat{\mu}_Z(\mathbf{X}_i) - \mu_Z(\mathbf{X}_i)\big),$$

we obtain

$$\begin{aligned} \widetilde{\mathbf{Y}}_i = A\widetilde{\mathbf{Z}}_i + \boldsymbol{\varepsilon}_i + A\big(\widehat{\mu}_Z(\mathbf{X}_i) - \mu_Z(\mathbf{X}_i)\big) \\ + \big(\mu_Y(\mathbf{X}_i) - \widehat{\mu}_Y(\mathbf{X}_i)\big). \end{aligned}$$

Define the remainder term

$$R_i = A\big(\widehat{\mu}_Z(\mathbf{X}_i) - \mu_Z(\mathbf{X}_i)\big) + \big(\mu_Y(\mathbf{X}_i) - \widehat{\mu}_Y(\mathbf{X}_i)\big).$$

Then

$$\widetilde{\mathbf{Y}}_i = A\widetilde{\mathbf{Z}}_i + \boldsymbol{\varepsilon}_i + R_i.$$

Substituting this representation into the estimator gives

$$\begin{aligned} \widehat{A} &= \left(\sum_{i=1}^{n} (A\widetilde{\mathbf{Z}}_i + \boldsymbol{\varepsilon}_i + R_i)\widetilde{\mathbf{Z}}_i^\top \right) \left(\sum_{i=1}^{n} \widetilde{\mathbf{Z}}_i \widetilde{\mathbf{Z}}_i^\top \right)^{-1} \\ &= A + \left(\sum_{i=1}^{n} \boldsymbol{\varepsilon}_i \widetilde{\mathbf{Z}}_i^\top \right) \left(\sum_{i=1}^{n} \widetilde{\mathbf{Z}}_i \widetilde{\mathbf{Z}}_i^\top \right)^{-1} \\ &\quad + \left(\sum_{i=1}^{n} R_i \widetilde{\mathbf{Z}}_i^\top \right) \left(\sum_{i=1}^{n} \widetilde{\mathbf{Z}}_i \widetilde{\mathbf{Z}}_i^\top \right)^{-1}. \end{aligned}$$

Hence,

$$\widehat{A} - A = \left(\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{\varepsilon}_i \widetilde{\mathbf{Z}}_i^\top\right)\left(\frac{1}{n}\sum_{i=1}^{n} \widetilde{\mathbf{Z}}_i \widetilde{\mathbf{Z}}_i^\top\right)^{-1} + \left(\frac{1}{n}\sum_{i=1}^{n} R_i \widetilde{\mathbf{Z}}_i^\top\right)\left(\frac{1}{n}\sum_{i=1}^{n} \widetilde{\mathbf{Z}}_i \widetilde{\mathbf{Z}}_i^\top\right)^{-1}. \tag{D.1}$$

We next show that the second term in (D.1) is asymptotically negligible. By Assumption 3,

$$\sup_{\mathbf{x}\in\mathcal{X}} \|\widehat{\mu}(\mathbf{x}) - \mu(\mathbf{x})\| = o_p(n^{-1/2}),$$

which implies

$$\max_{1\le i\le n} \|R_i\| = o_p(n^{-1/2}).$$

Since

$$\frac{1}{n}\sum_{i=1}^{n} \|\widetilde{\mathbf{Z}}_i\| = O_p(1),$$

it follows that

$$\left\|\frac{1}{n}\sum_{i=1}^{n} R_i \widetilde{\mathbf{Z}}_i^\top\right\| \le \max_{1\le i\le n} \|R_i\| \cdot \frac{1}{n}\sum_{i=1}^{n} \|\widetilde{\mathbf{Z}}_i\| = o_p(n^{-1/2}).$$

Moreover, by the law of large numbers,

$$\frac{1}{n}\sum_{i=1}^{n} \widetilde{\mathbf{Z}}_i \widetilde{\mathbf{Z}}_i^\top \xrightarrow{p} \Omega,$$

where

$$\Omega = \mathbb{E}\big[\widetilde{\mathbf{Z}}\widetilde{\mathbf{Z}}^\top\big].$$

Since Ω is positive definite by assumption,

$$\left(\frac{1}{n}\sum_{i=1}^{n} \widetilde{\mathbf{Z}}_i \widetilde{\mathbf{Z}}_i^\top\right)^{-1} \xrightarrow{p} \Omega^{-1}.$$

Therefore,

$$\widehat{A} - A = \Omega^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{\varepsilon}_i \widetilde{\mathbf{Z}}_i^\top\right) + o_p(n^{-1/2}).$$

Multiplying both sides by $\sqrt{n}$ yields

$$\sqrt{n}(\widehat{A} - A) = \Omega^{-1}\left(\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \boldsymbol{\varepsilon}_i \widetilde{\mathbf{Z}}_i^\top\right) + o_p(1).$$

Applying the vectorization operator,

$$\sqrt{n}\,\mathrm{vec}(\widehat{A} - A) = (\Omega^{-1} \otimes I_q)\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\mathrm{vec}\big(\boldsymbol{\varepsilon}_i\widetilde{\mathbf{Z}}_i^\top\big) + o_p(1).$$

Using the identity

$$\mathrm{vec}(ab^\top) = (b \otimes I_q)a,$$

we obtain

$$\mathrm{vec}\big(\boldsymbol{\varepsilon}_i\widetilde{\mathbf{Z}}_i^\top\big) = (\widetilde{\mathbf{Z}}_i \otimes I_q)\boldsymbol{\varepsilon}_i.$$

Define

$$U_i = (\widetilde{\mathbf{Z}}_i \otimes I_q)\boldsymbol{\varepsilon}_i.$$

Then $\{U_i\}$ are independent random vectors satisfying

$$\mathbb{E}(U_i) = 0,$$

because

$$\mathbb{E}(\boldsymbol{\varepsilon}_i \mid \mathbf{X}_i, \mathbf{Z}_i) = 0.$$

Further,

$$\begin{aligned}\mathrm{Cov}(U_i) &= \mathbb{E}\Big[(\widetilde{\mathbf{Z}}_i \otimes I_q)\Sigma(\mathbf{X}_i)(\widetilde{\mathbf{Z}}_i^\top \otimes I_q)\Big] \\ &= \mathbb{E}\Big[(\widetilde{\mathbf{Z}}_i\widetilde{\mathbf{Z}}_i^\top) \otimes \Sigma(\mathbf{X}_i)\Big] \\ &= \Xi.\end{aligned}$$

Since Assumptions 1 and 4 imply finite fourth moments, the multivariate Lindeberg–Feller central limit theorem applies, yielding

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n} U_i \xrightarrow{d} \mathcal{N}(0, \Xi).$$

Consequently,

$$\sqrt{n}\,\mathrm{vec}(\widehat{A} - A) \xrightarrow{d} \mathcal{N}\big(0, (\Omega^{-1} \otimes I_q)\Xi(\Omega^{-1} \otimes I_q)\big).$$

Absorbing the Kronecker notation into the matrix representation, we obtain

$$\sqrt{n}\,\mathrm{vec}(\widehat{A} - A) \xrightarrow{d} \mathcal{N}\big(0, \Omega^{-1}\Xi\Omega^{-1}\big).$$

This completes the proof.

□

E Proof of Theorem 5.4

Proof. The spline estimator admits the linear expansion

$$\widehat{m}(\mathbf{x}) - m(\mathbf{x}) = B(\mathbf{x})^\top \Gamma^{-1} \frac{1}{n} \sum_{i=1}^{n} B(\mathbf{X}_i)\boldsymbol{\varepsilon}_i + o_p(n^{-1/2}).$$

Hence

$$\sqrt{n}(\widehat{m}(\mathbf{x}) - m(\mathbf{x})) = B(\mathbf{x})^\top \Gamma^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} B(\mathbf{X}_i)\boldsymbol{\varepsilon}_i + o_p(1).$$

The class

$$\{B(\cdot)^\top \Gamma^{-1} v : \|v\| \le C\}$$

is finite-dimensional and therefore Donsker. By the multivariate functional central limit theorem,

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} B(\mathbf{X}_i)\boldsymbol{\varepsilon}_i \Rightarrow \mathcal{Z},$$

where $\mathcal{Z}$ is Gaussian with covariance operator Ω.

Tightness follows from finite dimensionality of the spline space. The covariance expression is obtained by direct calculation. □

F Proof of Theorem 5.6

Proof. Define the centered stochastic process

$$G_n(\mathbf{x}) = \sqrt{n}\big(\widehat{m}(\mathbf{x}) - m(\mathbf{x})\big), \qquad \mathbf{x} \in \mathcal{X}.$$

By Theorem 5.4,

$$G_n \Rightarrow \mathbb{G} \quad \text{in } \ell^\infty(\mathcal{X})^q,$$

where $\mathbb{G}$ is a mean-zero Gaussian process with covariance function

$$K(\mathbf{x}, \mathbf{x}') = \mathrm{Cov}\big(\mathbb{G}(\mathbf{x}), \mathbb{G}(\mathbf{x}')\big).$$

Recall that

$$T_n = \int_{\mathcal{X}} \|G_n(\mathbf{x})\|^2 w(\mathbf{x})\, d\mathbf{x},$$

where $w(\mathbf{x})$ is bounded and strictly positive on $\mathcal{X}$.

We first verify continuity of the functional

$$\Phi(g) = \int_{\mathcal{X}} \|g(\mathbf{x})\|^2 w(\mathbf{x})\, d\mathbf{x}, \qquad g \in \ell^\infty(\mathcal{X})^q.$$

Let $\{g_n\}$ be a sequence such that

$$\|g_n - g\|_\infty \to 0.$$

Since w is bounded on the compact set $\mathcal{X}$, there exists a constant $C > 0$ such that

$$0 < w(\mathbf{x}) \le C \qquad \text{for all } \mathbf{x} \in \mathcal{X}.$$

Then

$$\begin{aligned} |\Phi(g_n) - \Phi(g)| &= \left| \int_{\mathcal{X}} \Big(\|g_n(\mathbf{x})\|^2 - \|g(\mathbf{x})\|^2 \Big) w(\mathbf{x})\, d\mathbf{x} \right| \\ &\le C \int_{\mathcal{X}} \big| \|g_n(\mathbf{x})\|^2 - \|g(\mathbf{x})\|^2 \big|\, d\mathbf{x}. \end{aligned}$$

Using the identity

$$|a^2 - b^2| = |a - b||a + b|,$$

together with uniform boundedness of g_n and g on $\mathcal{X}$, we obtain

$$|\Phi(g_n) - \Phi(g)| \le C' \|g_n - g\|_\infty \to 0,$$

for some finite constant $C' > 0$. Hence Φ is continuous on $\ell^\infty(\mathcal{X})^q$.

Therefore, by the continuous mapping theorem,

$$T_n = \Phi(G_n) \Rightarrow \Phi(\mathbb{G}) = \int_{\mathcal{X}} \|\mathbb{G}(\mathbf{x})\|^2 w(\mathbf{x})\, d\mathbf{x}.$$

We now characterize the limiting distribution. Define the covariance operator

$$(\mathcal{K} f)(\mathbf{x}) = \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{x}') f(\mathbf{x}') w(\mathbf{x}')\, d\mathbf{x}',$$

acting on $L_2(\mathcal{X}, w)$.

Since K is symmetric and positive semidefinite, $\mathcal{K}$ is a compact self-adjoint operator. Hence, by the spectral theorem for compact operators, there exist nonnegative eigenvalues $\{\lambda_k\}_{k \ge 1}$ and orthonormal eigenfunctions $\{\phi_k\}_{k \ge 1}$ such that

$$K(\mathbf{x}, \mathbf{x}') = \sum_{k=1}^{\infty} \lambda_k \phi_k(\mathbf{x}) \phi_k(\mathbf{x}').$$

Consequently, the Gaussian process $\mathbb{G}$ admits the Karhunen–Loève expansion

$$\mathbb{G}(\mathbf{x}) = \sum_{k=1}^{\infty} \sqrt{\lambda_k}\, Z_k \phi_k(\mathbf{x}),$$

where $\{Z_k\}$ are independent standard normal random variables.

Substituting this representation into the limiting quadratic functional gives

$$\begin{aligned} \int_{\mathcal{X}} \|\mathbb{G}(\mathbf{x})\|^2 w(\mathbf{x})\, d\mathbf{x} &= \int_{\mathcal{X}} \left| \sum_{k=1}^{\infty} \sqrt{\lambda_k}\, Z_k \phi_k(\mathbf{x}) \right|^2 w(\mathbf{x})\, d\mathbf{x} \\ &= \sum_{k=1}^{\infty} \lambda_k Z_k^2, \end{aligned}$$

where the cross-product terms vanish due to the orthonormality of $\{\phi_k\}$ in $L_2(\mathcal{X}, w)$.

Therefore,

$$T_n \xrightarrow{d} \sum_{k=1}^{\infty} \lambda_k Z_k^2,$$

which completes the proof.

□

G Proof of Theorem 5.7

Proof. The model here is given by $Y_i = AZ_i + m(X_i) + \epsilon_i$, for $i = 1, 2, \cdots, n$, where A is a parameter, $m(X)$ is a nuisance function.
Also it is given that

$$\mu_Y(x) = E(Y|X = x) \text{ and } \mu_Z(x) = E(Z|X = x).$$

Considering the model we can have the following deductions:

$$\begin{aligned} Y_i &= AZ_i + m(X_i) + \epsilon_i \\ \implies E(Y_i|X = X_i) &= AE(Z_i|X = X_i) + m(X_i) + E(\epsilon_i|X = X_i) \\ \implies E(Y_i|X = X_i) &= AE(Z_i|X = X_i) + m(X_i) \\ \implies \mu_Y(x_i) &= A\mu_Z(X_i) + m(X_i). \end{aligned} \tag{G.1}$$

Therefore,

$$\begin{aligned} (Y_i - \mu_Y(X_i)) &= A(Z_i - \mu_z(X_i)) + \epsilon_i \\ i.e., \quad Y_i^* &= AZ_i^* + \epsilon_i, \end{aligned} \tag{G.2}$$

where $Y_i^* = (Y_i - \mu_Y(X_i))$ and $Z_i^* = (Z_i - \mu_z(X_i))$.
Also the residual variables are given by

$$\tilde{Y}_i = Y_i - \hat{\mu}_Y(X_i) \text{ and } \tilde{Z}_i = Z_i - \hat{\mu}_Z(X_i).$$

Thus,

$$\begin{aligned} \tilde{Y}_i &= Y_i - \hat{\mu}_Y(X_i) \\ &= Y_i - \mu_Y(X_i) + \mu_Y(X_i) - \hat{\mu}_Y(X_i) \\ &= Y_i^* - (\hat{\mu}_Y(X_i) - \mu_Y(X_i)) \\ &= Y_i^* - \delta_{Y,i}, \quad \text{where } \delta_{Y,i} = \hat{\mu}_Y(X_i) - \mu_Y(X_i). \end{aligned} \tag{G.3}$$

Similarly,

$$\tilde{Z}_i = Z_i^* - \delta_{Z,i}, \quad \text{where } \delta_{Z,i} = \hat{\mu}_Z(X_i) - \mu_Z(X_i). \tag{G.4}$$

Therefore from (G.2), using (G.3) and (G.4) we get that

$$\begin{aligned} Y_i^* &= AZ_i^* + \epsilon_i \\ \Rightarrow (\tilde{Y}_i + \delta_{Y,i}) &= A(\tilde{Z}_i + \delta_{Z,i}) + \epsilon_i \\ \Rightarrow \tilde{Y}_i &= A\tilde{Z}_i + (\epsilon_i + A\delta_{Z,i} - \delta_{Y,i}) \\ \therefore \tilde{Y}_i &= A\tilde{Z}_i + \epsilon_i + R_i, \end{aligned} \tag{G.5}$$

where $R_i = (A\delta_{Z,i} - \delta_{Y,i})$.
Now

$$\begin{aligned} & ||R_i|| \\ = \; & ||A\delta_{Z,i} - \delta_{Y,i}|| \\ \leq \; & ||A|| \cdot ||\delta_{Z,i}|| + ||\delta_{Y,i}|| \end{aligned} \tag{G.6}$$

Since in spline estimation, $\sup_{x \in X_i} ||\hat{\mu}(x) - \mu_Y(x)|| = o_p(n^{-\frac{1}{2}})$, then we can assert that $||\delta_{Y,i}|| \simeq o_p(n^{-\frac{1}{2}})$. By similar type of argument we also conclude that $||\delta_{Z,i}|| \simeq o_p(n^{-\frac{1}{2}})$.
Thus from (G.5) we get that

$$\tilde{Y}_i = A\tilde{Z}_i + \epsilon_i. \tag{G.7}$$

We now try to estimate the values of A by minimizing the following

$$\sum_{i=1}^{n} ||\tilde{Y}_i - A\tilde{Z}_i||^2.$$

For that we consider the equation

$$\frac{\partial}{\partial A} \left(\sum_{i=1}^{n} ||\tilde{Y}_i - A\tilde{Z}_i||^2 \right) = 0. \tag{G.8}$$

Again,

$$\begin{aligned} ||\tilde{Y}_i - A\tilde{Z}_i||^2 &= (\tilde{Y}_i - A\tilde{Z}_i)^T(\tilde{Y}_i - A\tilde{Z}_i) \\ &= \tilde{Y}_i^T\tilde{Y}_i - 2\tilde{Y}_i^T A\tilde{Z}_i + \tilde{Z}_i^T A^T A\tilde{Z}_i \end{aligned}$$

Consider the polynomial

$$Q(A) = \sum_{i=1}^{n} \{\tilde{Y}_i^T\tilde{Y}_i - 2\tilde{Y}_i^T A\tilde{Z}_i + \tilde{Z}_i^T A^T A\tilde{Z}_i\}$$

Differentiating term by term we get that

$$\frac{\partial}{\partial A}\left(Q(A)\right) = \sum_{i=1}^{n} \frac{\partial}{\partial A}\left(\tilde{Y}_i^T\tilde{Y}_i\right) - 2\sum_{i=1}^{n} \frac{\partial}{\partial A}\left(\tilde{Y}_i^T A\tilde{Z}_i\right) + \sum_{i=1}^{n} \frac{\partial}{\partial A}\left(\tilde{Z}_i^T A^T A\tilde{Z}_i\right) \tag{G.9}$$

Since the first term is independent of A and by standard calculations we see that

$$\sum_{i=1}^{n} \frac{\partial}{\partial A}\left(\tilde{Y}_i^T\tilde{Y}_i\right) = 0$$

and

$$2\sum_{i=1}^{n} \frac{\partial}{\partial A}\left(\tilde{Y}_i^T A\tilde{Z}_i\right) = 2\sum_{i=1}^{n}\left(\tilde{Y}_i\tilde{Z}_i^T\right)$$

Again

$$(\tilde{Z}_i^T A^T A \tilde{Z}_i) \;=\; \text{trace}\left(A \tilde{Z}_i \tilde{Z}_i^T A^T\right)$$

Therefore,

$$\begin{aligned}
& \sum_{i=1}^{n} \frac{\partial}{\partial A}(\tilde{Z}_i^T A^T A \tilde{Z}_i) \\
= \;& \frac{\partial}{\partial A} \sum_{i=1}^{n} (\tilde{Z}_i^T A^T A \tilde{Z}_i) \\
= \;& \frac{\partial}{\partial A} \sum_{i=1}^{n} \text{trace}\left(A \tilde{Z}_i \tilde{Z}_i^T A^T\right) \\
= \;& \frac{\partial}{\partial A} A \sum_{i=1}^{n} \left[\text{trace}\left(\tilde{Z}_i \tilde{Z}_i^T\right)\right] A^T \\
= \;& \frac{\partial}{\partial A}[AMA^T], \quad \text{where} \quad M = \sum_{i=1}^{n} \left[\text{trace}\left(\tilde{Z}_i \tilde{Z}_i^T\right)\right] \\
= \;& 2A \sum_{i=1}^{n} \left(\tilde{Z}_i \tilde{Z}_i^T\right)
\end{aligned}$$

Therefore from (G.9) we have

$$\begin{aligned}
-2 \sum_{i=1}^{n} \left(\tilde{Y}_i \tilde{Z}_i^T\right) \;&+\; 2A \sum_{i=1}^{n} \left(\tilde{Z}_i \tilde{Z}_i^T\right) = 0 \\
\Rightarrow \sum_{i=1}^{n} \left(\tilde{Y}_i \tilde{Z}_i^T\right) \;&=\; A \sum_{i=1}^{n} \left(\tilde{Z}_i \tilde{Z}_i^T\right)
\end{aligned}$$

Therefore, the least square estimator of A is given by

$$\hat{A} = \left(\sum_{i=1}^{n} \tilde{Y}_i \tilde{Z}_i^T\right) \left(\sum_{i=1}^{n} \tilde{Z}_i \tilde{Z}_i^T\right)^{-1}. \tag{G.10}$$

Keeping (G.2), (G.3) and (G.4) in mind, we see that the numerator of (G.10) can be expressed as follows:

$$\begin{aligned}
\sum_{i=1}^{n} \tilde{Y}_i \tilde{Z}_i^T \;&=\; \sum_{i=1}^{n} [A(Z_i - \mu_Z(X_i)) + \epsilon_i - \delta_{Y,i}][(Z_i - \mu_Z(X_i)) - \delta_{Z,i}]^T \\
&= \sum_{i=1}^{n} A(Z_i - \mu_Z(X_i))(Z_i - \mu_Z(X_i))^T + \sum_{i=1}^{n} \epsilon_i (Z_i - \mu_Z(X_i))^T + \sum_{i=1}^{n} \delta_{Y,i} \delta_{Z,i}^T \\
&\quad - \sum_{i=1}^{n} \delta_{Y,i} (Z_i - \mu_Z(X_i))^T - \sum_{i=1}^{n} A(Z_i - \mu_Z(X_i)) \delta_{Z,i}^T - \sum_{i=1}^{n} \epsilon_i \delta_{Z,i}^T
\end{aligned} \tag{G.11}$$

Since we know that in spline estimation,

$$\sup_x ||\hat{\mu}(x) - \mu(x)|| = o_p(n^{-\frac{1}{2}}),$$

under the conditions $K_n^{-r} = o(n^{-\frac{1}{2}})$ and $\frac{K_n}{n} \to 0$, then it is easy to see that

$$\max_i ||\delta_{Y,i}|| = o_p(n^{-\frac{1}{2}})$$

and

$$\max_i ||\delta_{Z,i}|| = o_p(n^{-\frac{1}{2}}).$$

Consequently, we see that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} A(Z_i - \mu_Z(X_i))\delta_{Z,i}^T = o_p(1).$$

In the similar manner we see that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \delta_{Y,i}(Z_i - \mu_Z(X_i))^T = o_p(1)$$

and

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \delta_{Y,i}\delta_{Z,i}^T = o_p(1).$$

Therefore, from (G.11) we see that

$$\begin{aligned} \frac{1}{n}\sum_{i=1}^{n} \tilde{Y}_i \tilde{Z}_i^T &= \frac{A}{n}\sum_{i=1}^{n}(Z_i - \mu_Z(X_i))(Z_i - \mu_Z(X_i))^T + \frac{1}{n}\sum_{i=1}^{n}\epsilon_i(Z_i - \mu_Z(X_i))^T \\ &\quad + o_p(n^{-\frac{1}{2}}) \end{aligned} \tag{G.12}$$

Also from the denominator of the expression $\hat{A}$,

$$\frac{1}{n}\sum_{i=1}^{n} \tilde{Z}_i \tilde{Z}_i^T = \frac{1}{n}\sum_{i=1}^{n}(Z_i - \mu_Z(X_i))(Z_i - \mu_Z(X_i))^T + o_p(1)$$

Thus,

$$\frac{1}{n}\sum_{i=1}^{n} \tilde{Z}_i \tilde{Z}_i^T \to \Omega \tag{G.13}$$

Therefore from the expression of $\hat{A}$, using (G.12) we see that

$$\begin{aligned}
\hat{A} &= \left[A\left(\frac{1}{n}\sum_{i=1}^{n}\tilde{Z}_i\tilde{Z}_i^T\right)+\frac{1}{n}\sum_{i=1}^{n}\epsilon_i\tilde{Z}_i^T+o_p(n^{-\frac{1}{2}})\right]\left(\frac{1}{n}\sum_{i=1}^{n}\tilde{Z}_i\tilde{Z}_i^T\right)^{-1} \\
\Rightarrow \hat{A} &= A+\left(\frac{1}{n}\sum_{i=1}^{n}\epsilon_i\tilde{Z}_i^T\right)\left(\frac{1}{n}\sum_{i=1}^{n}\tilde{Z}_i\tilde{Z}_i^T\right)^{-1}+o_p(n^{-\frac{1}{2}}) \\
\Rightarrow (\hat{A}-A) &= \left(\frac{1}{n}\sum_{i=1}^{n}\epsilon_i\tilde{Z}_i^T\right)\left(\frac{1}{n}\sum_{i=1}^{n}\tilde{Z}_i\tilde{Z}_i^T\right)^{-1}+o_p(n^{-\frac{1}{2}}) \\
\Rightarrow \sqrt{n}(\hat{A}-A) &= \left(\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\epsilon_i\tilde{Z}_i^T\right)\left(\frac{1}{n}\sum_{i=1}^{n}\tilde{Z}_i\tilde{Z}_i^T\right)^{-1}+o_p(1) \qquad \text{(G.14)}
\end{aligned}$$

Thus by (G.12) we have

$$\begin{aligned}
\sqrt{n}(\hat{A}-A) &= \Omega^{-1}\left(\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\epsilon_i\tilde{Z}_i^T\right)+o_p(1) \\
&= \Omega^{-1}\left(\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\epsilon_i(Z_i-\mu_Z(X_i))^T\right)+o_p(1)
\end{aligned}$$

Consequently, as per Theorem 5.2, it is concluded that

$$\sqrt{n}\ \mathrm{vec}(\hat{A}-A) = \Omega^{-1}\left(\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\mathrm{vec}\left[\epsilon_i(Z_i-\mu_Z(X_i))^T\right]\right)+o_p(1).$$

□